An odd-parameter factor exclusion for the second cuboid quintic

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Abstract

We give a Lean-checked proof that the normalized second cuboid quintic has no rational divisor of degree two when its parameter is the square of the ratio of two positive unequal odd integers. The proof transports factor degrees through two finite-dimensional quotient-algebra isomorphisms, then uses an elementary obstruction modulo four. It includes reducible and repeated quadratic divisors and does not require coprimality of the parameters. This is a research note on one parameter restriction. The opposite-parity case and the full second cuboid conjecture remain unresolved here. The originality of the restriction has not been established.

1 Statement and scope

For $s \in \mathbb{Q}$, define

$$\begin{aligned} P_s(x) = x^5 &+ (2+s)(3-2s)x^4 \\ &+ (1+10s+4s^2-14s^3+s^4)x^3 \\ &- s(1-14s+4s^2+10s^3+s^4)x^2 \\ &- s^3(1+2s)(-2+3s)x - s^5. \end{aligned} \tag{1}$$

This is the normalization used in the recent quadratic-factor study [3]. For Sharipov's even degree-ten polynomial,

$$Q_{p,q}(t) = q^{20} P_{(p/q)^2}(t^2/q^4).$$

Our formal statement concerns P_s itself; the displayed relation explains its role in the factorization problem.

Theorem 1. *Let $p, q \in \mathbb{Z}$ be positive, odd, and unequal, and put $s = (p/q)^2$. There is no polynomial $D \in \mathbb{Q}[x]$ with $\deg D = 2$ and $D \mid P_s$.*

Coprimality is unnecessary. No irreducibility or squarefreeness assumption is imposed on D. The theorem proves a restriction within the quadratic-factor problem; it does not prove irreducibility of P_s or $Q_{p,q}$. For coprime positive parameters, the unhandled parity class has one parameter even and the other odd. The condition $p \neq q$ is essential to the changes of variable: $P_1(x) = (x-1)(x+1)^4$.

2 Two changes of variable

Throughout this section assume $s > 0$ and $s \neq 1$, and set

$$h = \frac{s-1}{s+1}, \qquad k = h^2.$$

Then $h \neq 0$ and $0 < k < 1$. Introduce the monic quintics

$$T_k(w) = w(w^2-1)^2 + 4(1-k)(w^2-k),$$
$$g_k(v) = v^5 + v^4 - 2v^3 - 2v^2 + 2 - 2k.$$

Lemma 2. *There are $\mathbb{Q}$-algebra isomorphisms*

$$\mathbb{Q}[x]/(P_s) \;\cong\; \mathbb{Q}[w]/(T_k) \;\cong\; \mathbb{Q}[v]/(g_k).$$

Proof. All three algebras have dimension five. In $A = \mathbb{Q}[x]/(P_s)$, the element $x+s$ is a unit because

$$P_s(-s) = -2s^3(s-1)^4 \neq 0.$$

Define $w = h(x-s)/(x+s)$, with division interpreted as multiplication by the inverse of a unit. The polynomial identity

$$(h-w)^5 P_s\left(s\frac{h+w}{h-w}\right) = 2s^3(s-1)^4 T_{h^2}(w) \tag{2}$$

holds wherever the displayed denominators are defined. Clearing denominators also gives an identity in the quotient algebra, so $T_k(w) = 0$ in A. Here $h-w = 2hs/(x+s)$ is a unit. The inverse expression

$$x = s\frac{h+w}{h-w}$$

shows that w generates A. To justify the use of the inverse within the subalgebra generated by w, use the following elementary fact: in a finite-dimensional commutative algebra over a field, a subalgebra containing an ambient unit also contains its inverse. Multiplication by that unit is injective on the subalgebra, hence surjective. Thus the homomorphism from $\mathbb{Q}[w]/(T_k)$ to A is surjective. Equal dimensions make it an isomorphism.

For the second isomorphism, work in $B = \mathbb{Q}[v]/(g_k)$ and set $w = 1 - v^2$. Direct polynomial identities give

$$T_k(1-v^2) = g_k(v)g_k(-v), \tag{3}$$
$$g_k(v) = v^2(v^2-2)(v+1) + 2(1-k), \tag{4}$$
$$2k - 1 - (1-v^2)^2 - v((1-v^2)^2 - 1) = -g_k(v). \tag{5}$$

Equation (3) gives $T_k(w) = 0$ in B. By (4), the element $w^2 - 1 = v^2(v^2-2)$ is a unit, since its product with $v+1$ is $-2(1-k) \neq 0$. Equation (5) therefore yields

$$v = \frac{2k - 1 - w^2}{w^2 - 1}.$$

The inverse-in-subalgebra argument again proves that w generates B. The resulting surjection $\mathbb{Q}[w]/(T_k) \longrightarrow B$ is an isomorphism. □

Lemma 3. *If P_s has a degree-two divisor over $\mathbb{Q}$, then g_k has a degree-two divisor over $\mathbb{Q}$.*

Proof. Let $D \mid P_s$ with $\deg D = 2$. Composing the isomorphisms of Lemma 2 with the quotient onto $\mathbb{Q}[x]/(D)$ gives a surjection from $\mathbb{Q}[v]$ to a two-dimensional $\mathbb{Q}$-algebra, whose kernel contains g_k. Its kernel is a principal ideal (E), and $\dim_{\mathbb{Q}} \mathbb{Q}[v]/(E) = \deg E = 2$. Hence $E \mid g_k$ with $\deg E = 2$. This argument also applies when D splits or has a repeated root. □

3 The integral obstruction

For positive odd p, q, put

$$d = \frac{p^2 + q^2}{2} \in \mathbb{Z}.$$

Odd squares are 1 modulo 4, so d is odd. Moreover

$$k = \left(\frac{p^2 - q^2}{p^2 + q^2}\right)^2, \qquad 1 - k = \frac{p^2 q^2}{d^2}.$$

Multiplication by d^2 gives the integral quintic

$$F(v) = d^2 g_k(v) = d^2(v^5 + v^4 - 2v^3 - 2v^2) + 2p^2q^2. \tag{6}$$

Lemma 4. *The polynomial F in (6) has no degree-two divisor over $\mathbb{Q}$.*

Proof. Suppose it has one. Gauss's lemma gives a factorization $F = AB$ in $\mathbb{Z}[v]$ with $\deg A = 2$ and $\deg B = 3$. There is no primitivity assumption on F: clear denominators in a rational divisor, take its primitive part, and apply Gauss's lemma to this primitive divisor. The complementary integral factor absorbs any content of F.

The leading coefficients of A and B multiply to d^2, which is odd. Their reductions modulo two therefore retain degrees two and three. But in $\mathbb{F}_2[v]$,

$$\overline{F} = v^4(v + 1).$$

Any divisor of $v^4(v + 1)$ with nonzero constant coefficient is coprime to v^4, so it divides $v + 1$ and has degree at most one. Consequently both $\overline{A}$ and $\overline{B}$ have zero constant coefficient. Thus $A(0)$ and $B(0)$ are even, forcing $4 \mid F(0)$. This contradicts

$$F(0) = 2p^2q^2 \equiv 2 \pmod 4. \qquad \square$$

Proof of Theorem 1. Positivity and $p \neq q$ give $s > 0$ and $s \neq 1$. A degree-two divisor of P_s would give one of g_k by Lemma 3, and hence one of $F = d^2 g_k$. Lemma 4 rules this out. $\square$

4 Formal verification and remaining work

The final declaration is

```
NumberTheoryResearch.BatchA.Cuboid.
    odd_odd_P_no_quadratic.
```

Its parameters are $p, q : \mathbb{Z}$, proofs that they are odd, positive, and unequal, and $D : \mathbb{Q}[X]$ with natural degree two. Its conclusion is the negation of $D \mid P_{(p/q)^2}$. The use of natural degree adds no zero polynomial exception, because the zero polynomial has natural degree zero.

The sources use Lean 4.35.0 and mathlib commit

```
e91b128082cc221d6432a6a5aadcd3243af8e76c.
```

The transitive axiom report for the theorem is exactly

```
propext,  Classical.choice,  Quot.sound.
```

No admitted proof, custom axiom, or native evaluation axiom is used. From the repository root, a fresh compilation and axiom audit runs with

```
scripts/with_lean.sh lake env python3
research/parent/cuboid-formal/verify.py
```

The command is a single shell line. The verifier compiles all eight source modules into a new temporary directory, so it does not rely on previously compiled research files. The public source repository is GPT-6 Astra Number Theory.

The parity argument is a classical local factor-degree obstruction. Focused searches and inspection of the cited cuboid papers did not identify this exact odd/odd restriction, but that does not establish originality. The project therefore retains it as a verified research restriction with novelty unverified, rather than counting it as a solved open problem. The recent quadratic-factor paper [3] gives computational evidence and a conditional conclusion for the full parameter range. The present proof removes the odd/odd case from our remaining target; it supplies no conclusion for the opposite-parity case.

References

[1] R. Sharipov, *Perfect cuboids and irreducible polynomials*, arXiv:1108.5348v1 (2011), https://arxiv.org/abs/1108.5348v1.

[2] R. Sharipov, *A note on the second cuboid conjecture. Part I*, arXiv:1201.1229v1 (2012), https://arxiv.org/abs/1201.1229v1.

[3] V. Asiryan and R. L. Rathbun, *Computational Evidence Against Quadratic-Cubic Factorization for the Second Cuboid Quintic*, arXiv:2601.07899v2 (2026), https://arxiv.org/abs/2601.07899v2.